DATE : 23.10.2026

DT/NT : DT

LESSON : AWS

SUBJECT: EFS

BATCH : B 424-020DE

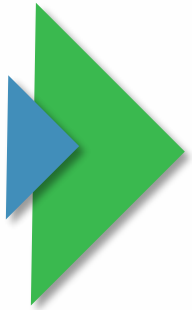
AWS-DEVOPS

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- Features of EFS
- Comparison of Storage Systems

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Introduction to EFS

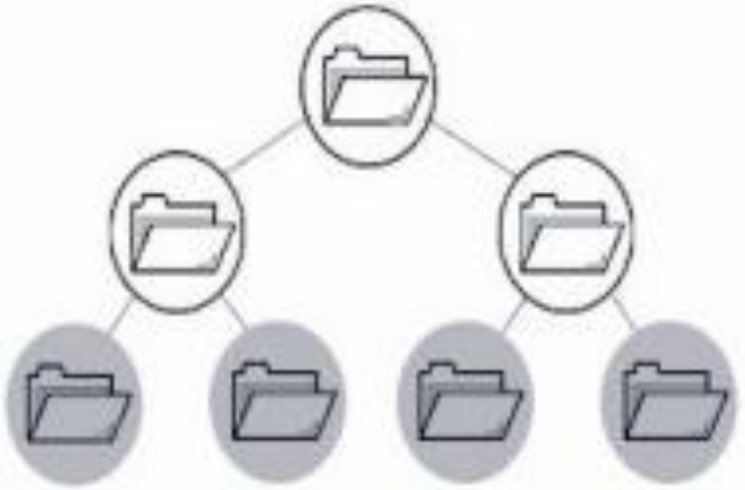
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Introduction to EFS

What is EFS?

EFS provides simple, scalable, and elastic file storage for use with AWS Cloud services and on-premises resources.





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Introduction to EFS

What is EFS?

Amazon Elastic File System (EFS) is a serverless, fully elastic file storage solution designed to help you share file data without the need to provision or manage storage capacity and performance.

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Features of EFS

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Features of EFS

Scalability-Cost

- Since EFS is scalable, it automatically increases and decreases storage capacity as you add or delete files.
- There are no minimum fees or setup costs.



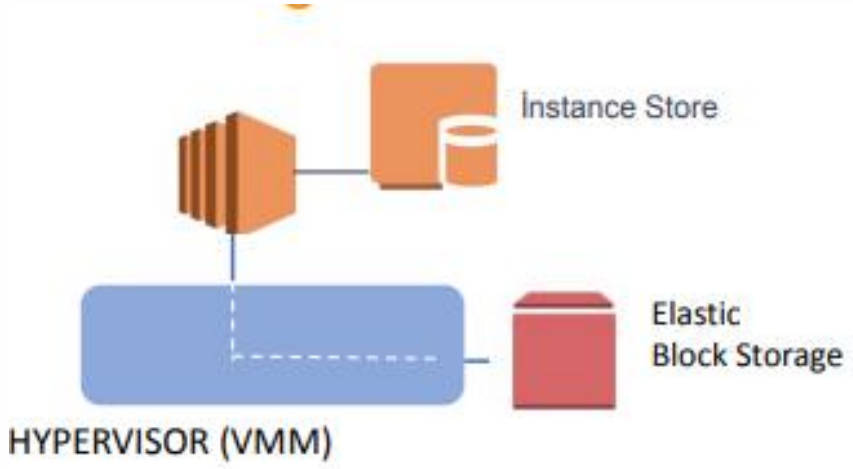


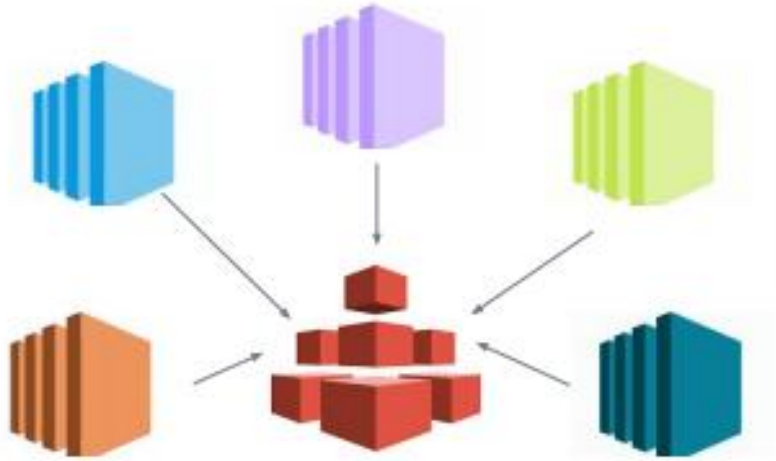
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Features of EFS

Attaching

- Unlike EBS, multiple Amazon EC2 instances (Linux only) can simultaneously access an Amazon EFS file system, even across different Availability Zones (AZs).

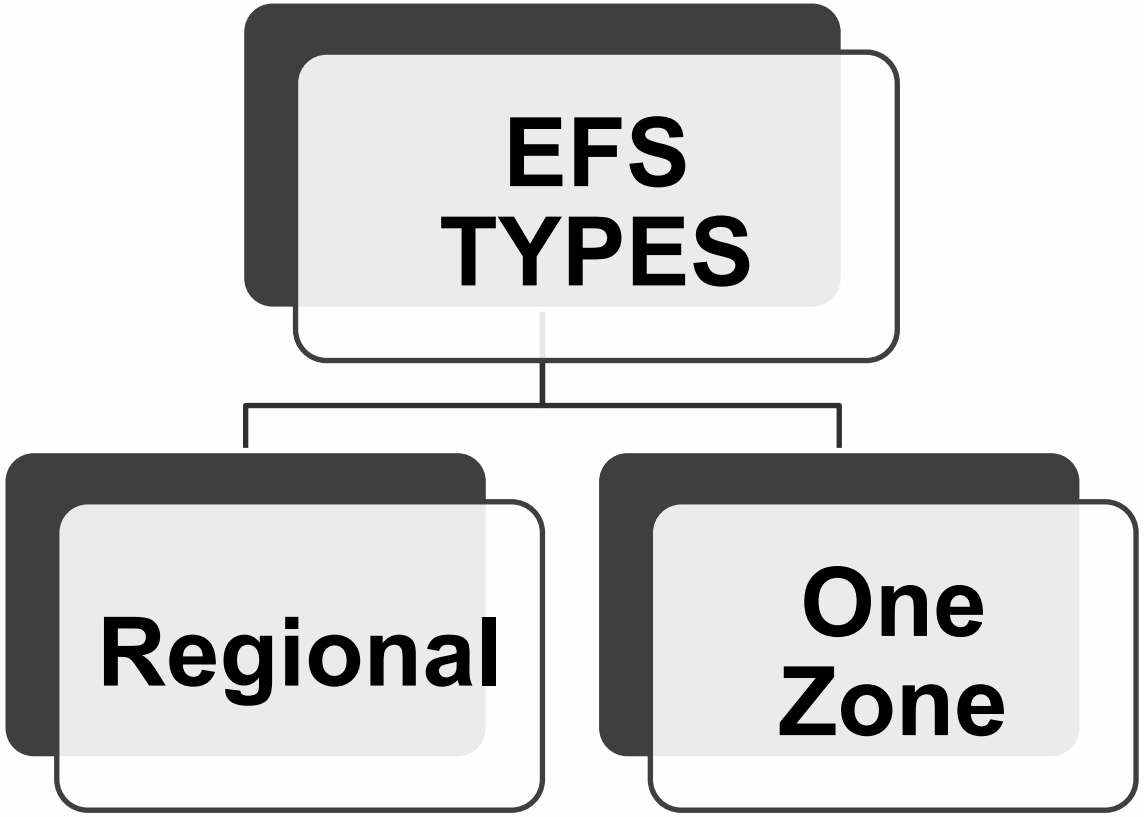




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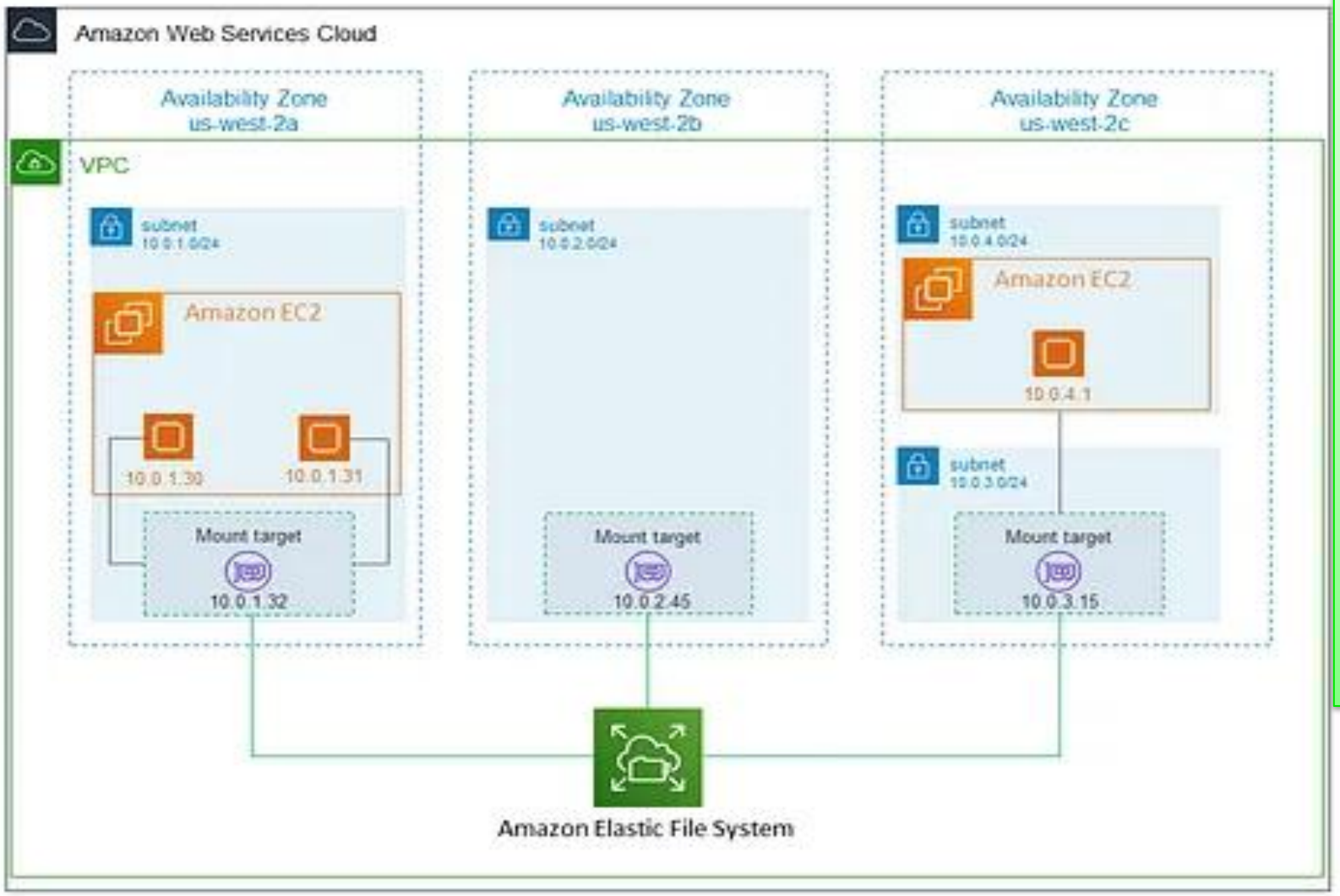
Features of EFS

EFS file system types



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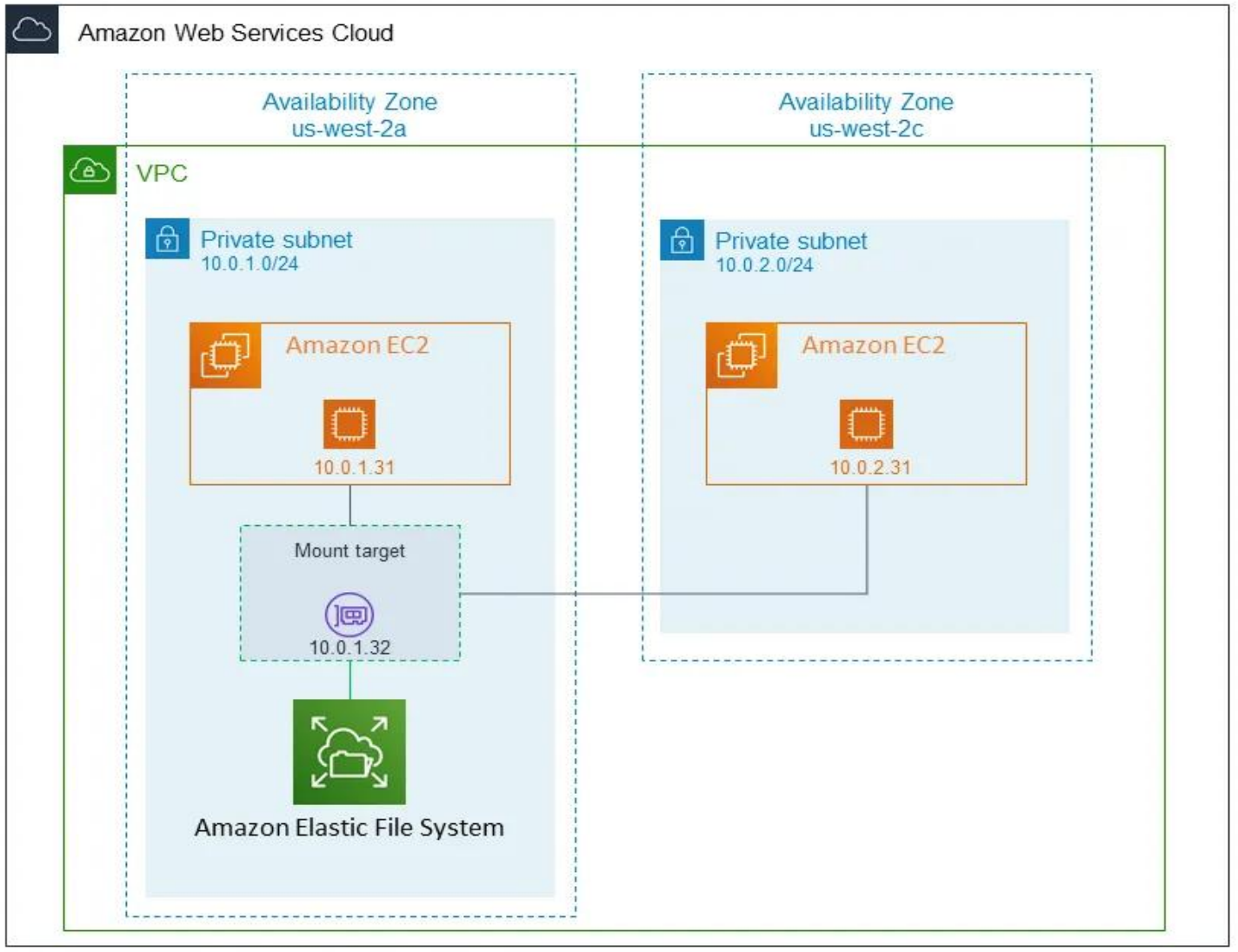
EFS Structure : Mount Target (for Regional Storage Class)



- A Mount Target is an AZ-based component.
- You can create only one Mount Target per Availability Zone (AZ).

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EFS Structure : Mount Target (for One-Zone Storage Class)

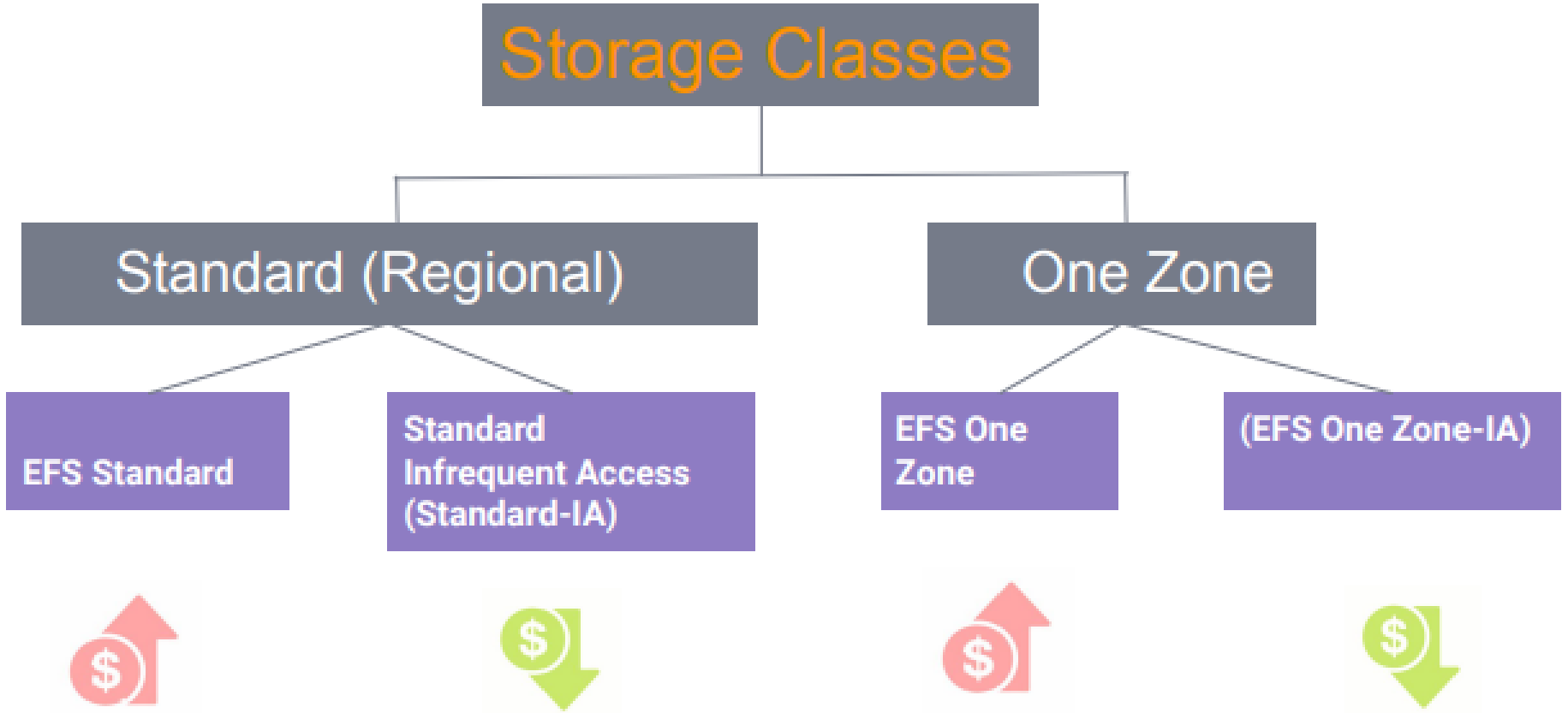


- A Mount Target is created in only one subnet within its respective AZ.
- Other AZs communicate with EFS using this Mount Target.

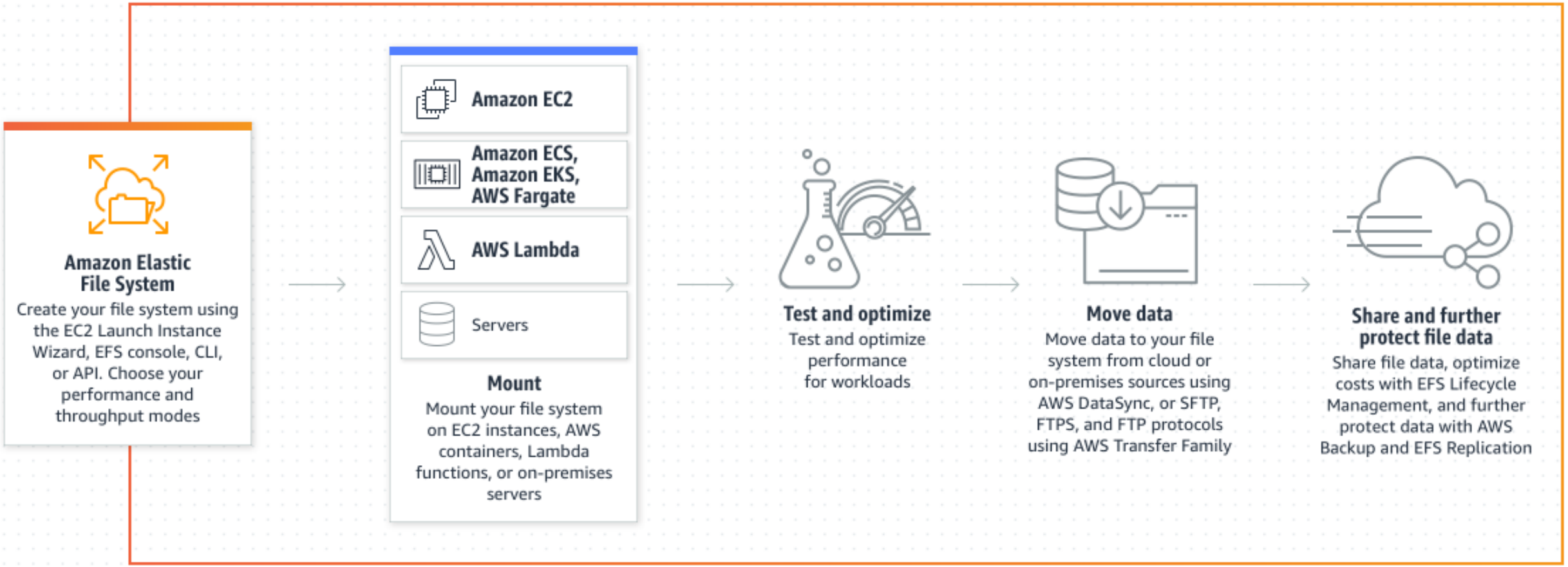
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Features of EFS

Storage Classes



Features of EFS



Comparison of Storage Systems

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Comparison of Storage Systems



Amazon EFS

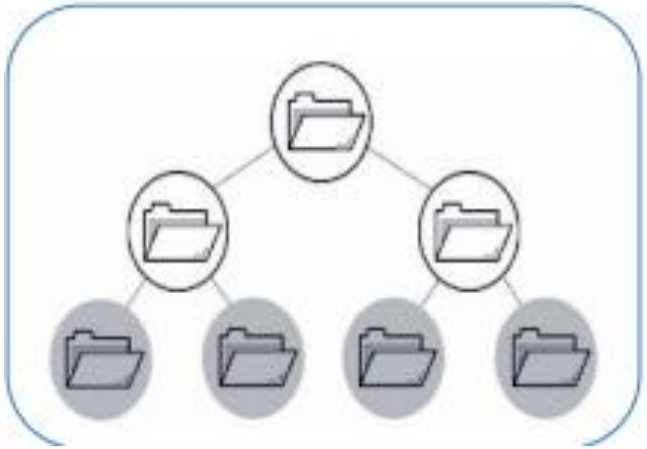


S3



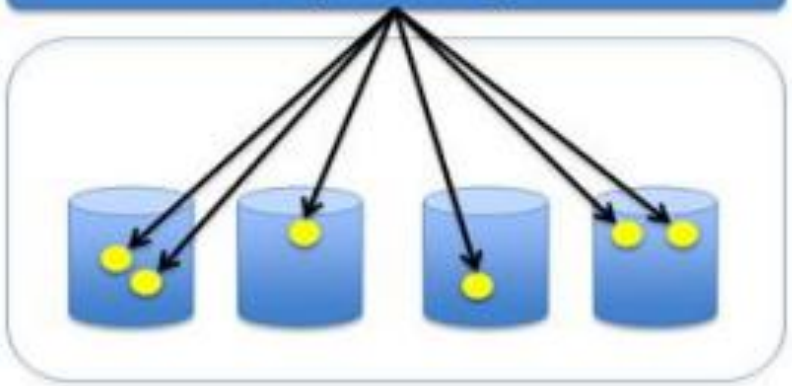
Amazon Elastic Block Storage (EBS)

File Storage

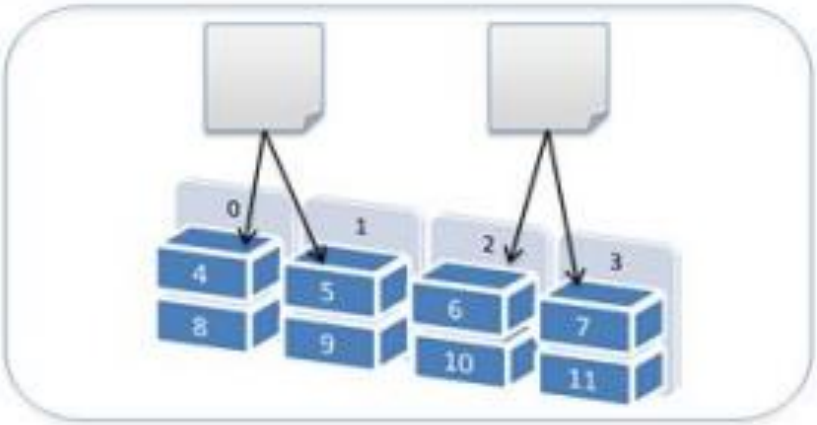


HTTP(S) Interface

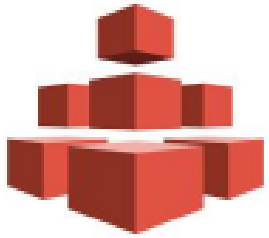
Object Storage



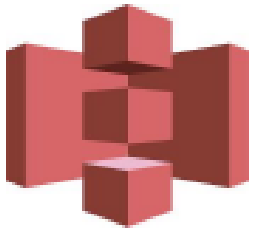
Block Storage



Comparison of Storage Systems



Amazon EFS



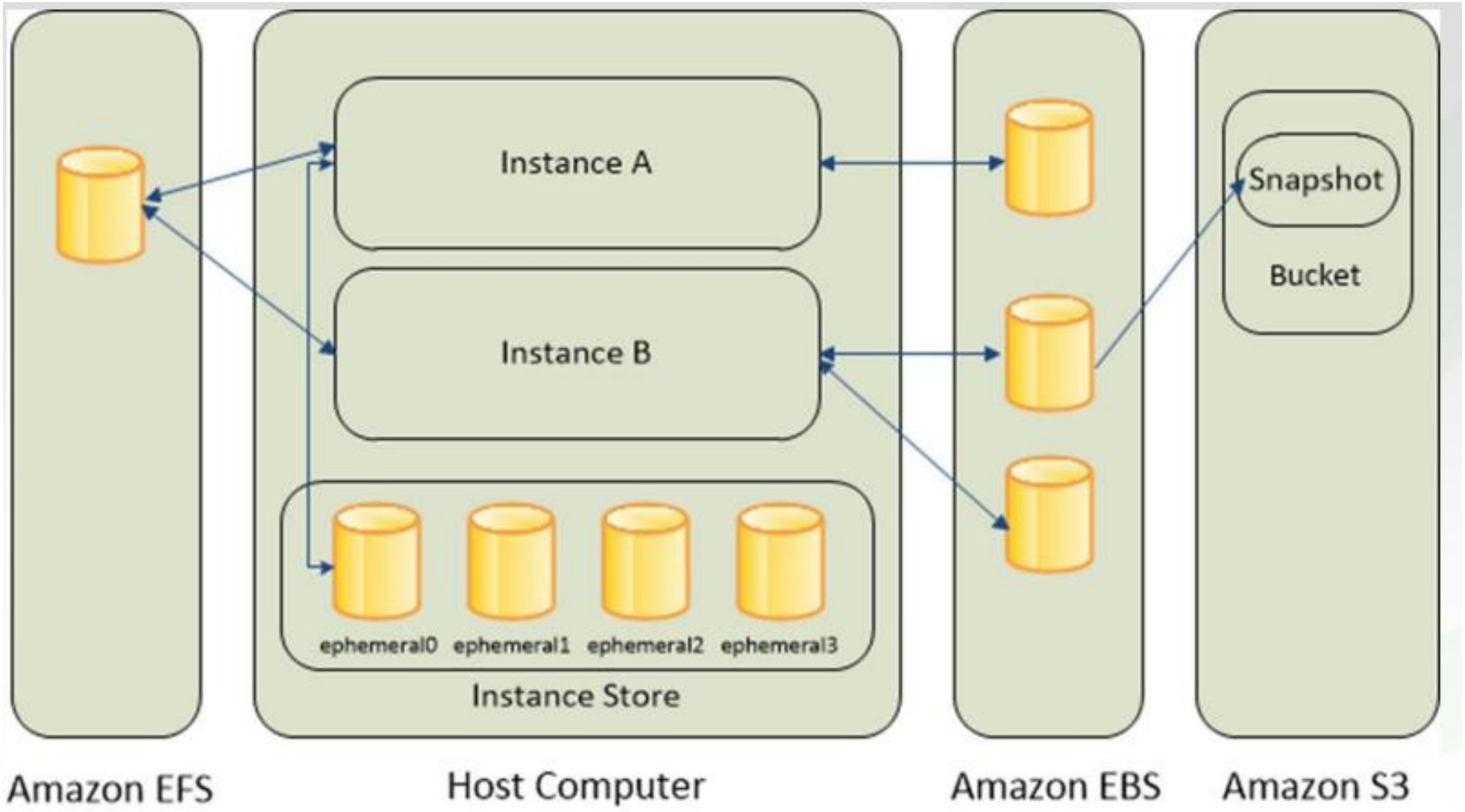
S3



EBS

- Cost Optimized : S3 > EBS > EFS
- Speed : EBS , EFS >S3
- EC2 mount : S3 : No
  EBS : Single*
  EFS : Multiple
- Storage Capacity : S3, EFS = ∞ vs. EBS **=64 TiB**

Comparison of Storage Systems

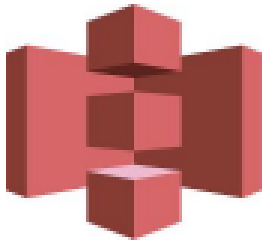


Comparison of Storage Systems



Amazon EFS

- Multi-client access
- Scalable applications
- Applications requiring high availability



S3

- Data backup and archiving
- Website assets
- Big data storage
- Application data



EBS

- Databases
- Boot volume
- Workloads requiring high performance

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Comparison of Storage Systems

Block Based

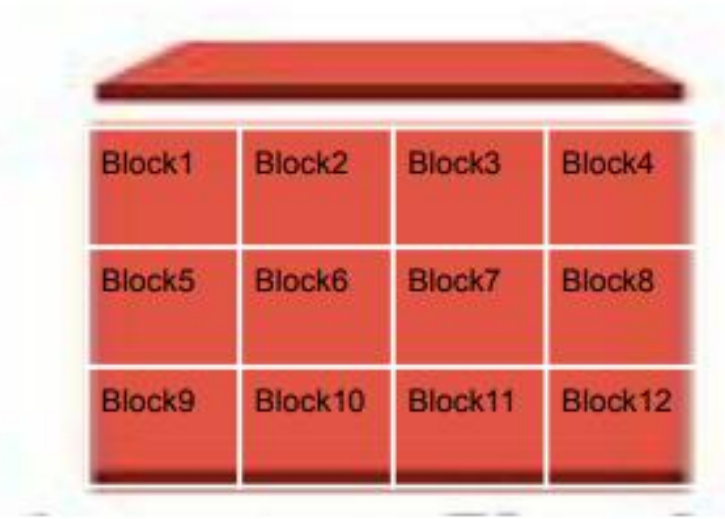


Totally = 16KB in size

Divide your object into the blocks max 4KB inseize

Each Block=4KB in size

EBS



Each Block= 4KB in seize

Total blocks number= 12

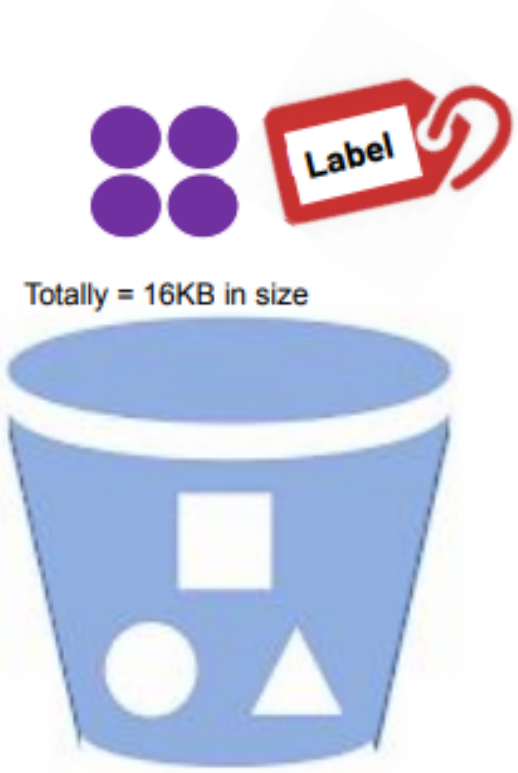
Totally = 48KB

Who can call the data? =Only related EC2

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Comparison of Storage Systems

Object Based



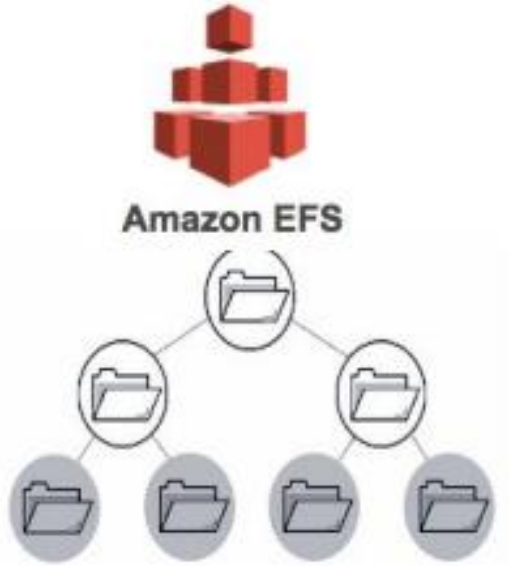
Who can call the data? = Anybody with public **internet** and permission

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Comparison of Storage Systems

File Based

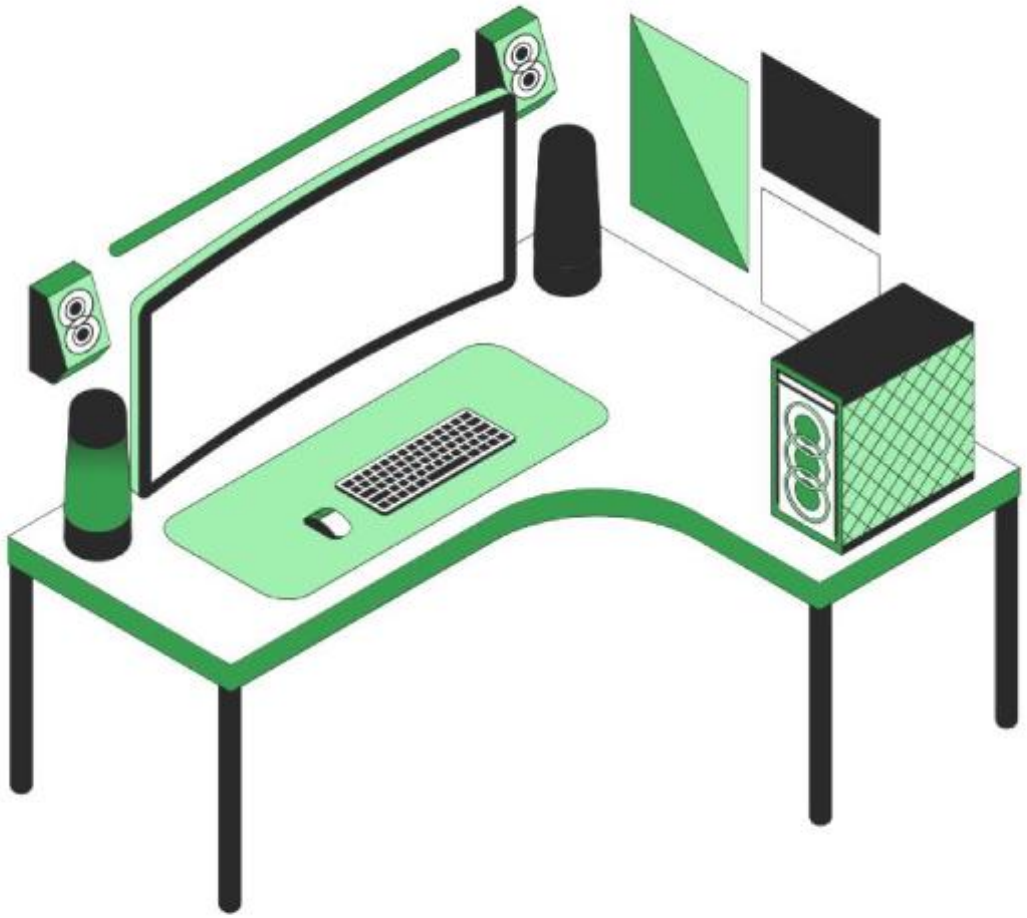




~/efs/techpro

Who can call the data? = Multiple or single EC2

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Do you have any questions?

Send it to us! We hope you learned something new.

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